ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PAPER 1 · PRACTICE D

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 8 year: frame, shell and solid structures, cantilevers, the five forces, gussets and corrugation, a design brief for a phone stand, material properties, ferrous and non-ferrous metals, rust prevention, recycling codes, a gear train with an idler gear, worm gears and cams, a two-door OR-logic alarm, switches and fuses, and the cost of electricity.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings and tables.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A balcony that is supported at only one end is a …

A cantilever. B arch. C truss. D column.

1.2 The force that tries to twist an object is …

A tension. B compression. C torsion. D shear.

1.3 Which of these is a solid structure?

A An egg shell B A dam wall C A bicycle frame D A tent

1.4 A metal that contains iron is a …

A non-ferrous metal. B precious metal.
C plastic. D ferrous metal.

1.5 The ability of a material to return to its original shape after it is stretched is called …

A hardness. B elasticity. C brittleness. D conductivity.

1.6 In a gear train, a small gear placed between the driver and the driven gear to keep the direction the same is an …

A idler gear. B rack. C cam. D sprocket.

1.7 Which mechanism changes rotary motion into up-and-down (reciprocating) motion?

A Belt drive B Cam and follower C Chain drive D Pulley

1.8 A doorbell that rings when EITHER the front OR the back door button is pressed uses …

A AND logic. B NOT logic. C OR logic. D series logic.

1.9 A fuse protects a circuit by …

A storing extra energy. B increasing the voltage.
C melting when the current is too high. D making the current flow faster.

1.10 Electricity used in a home is measured in …

A kilowatt-hours. B volts. C amperes. D ohms.

[10]

QUESTION 2: STRUCTURES

2.1 Name the type of structure (frame, shell or solid) for EACH of the following: (a) a pylon; (b) a cooldrink can; (c) a brick wall. (3)

2.2 Name the force (tension, compression, torsion, shear or bending) that acts in EACH of the following: (a) the rope of a swing; (b) the legs of a table; (c) a screwdriver turning a screw; (d) scissors cutting paper. (4)

2.3 Explain how EACH of the following makes a structure stronger: (a) a gusset; (b) corrugation (as in a corrugated iron roof). (4)

2.4 A learner wants to design a stand that holds a cell phone upright on a desk so that videos can be watched.

2.4.1 Write a design brief for the phone stand. (2)

2.4.2 Give THREE specifications for the stand. (3)

2.4.3 Give TWO ways to make the stand stable so that it does not fall over. (2)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Explain what EACH property means: (a) hardness; (b) strength; (c) brittleness. (3)

3.2 Give TWO examples of ferrous metals and TWO examples of non-ferrous metals. (4)

3.3 (a) What causes iron and steel to rust? (b) Give TWO ways to stop a steel gate from rusting. (4)

3.4 Plastic containers often have a triangle made of arrows with a number inside it. What does this symbol show, and why is it useful? (2)

3.5 Why are cooking pots often made of aluminium or stainless steel with plastic handles? (2)

[15]

QUESTION 4: MECHANICAL SYSTEMS

4.1 Study the gear train: the driver gear A has 12 teeth, idler gear B has 20 teeth, and the driven gear C has 36 teeth. Gear A turns at 90 r/min clockwise.

4.1.1 Calculate the gear ratio between A and C. (2)

4.1.2 Calculate the speed of gear C. (2)

4.1.3 In which direction does gear C turn? (1)

4.1.4 Does the idler gear change the speed of gear C? Explain. (2)

4.1.5 Is this gear train used to increase speed or to increase force (torque)? Explain. (2)

4.2 Give ONE use of a worm gear and ONE advantage of it. (2)

4.3 Draw a neat, labelled sketch of a cam and follower, and show the type of motion of each part. (4)

4.4 Explain why the moving parts of machines must be lubricated. (2)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 A shop wants a buzzer to sound when the front door OR the back door is opened. Each door has a switch that closes when the door opens.

5.1.1 Draw a circuit diagram with a battery, the two switches and the buzzer. (4)

5.1.2 Draw and complete a truth table for this system. (4)

5.2 Name TWO types of switches and give ONE use of each. (2)

5.3 A 2 kW kettle is used for 30 minutes every day. Electricity costs R3,50 per kWh. Calculate the cost of using the kettle for 30 days. Use: energy (kWh) = power (kW) × time (h). (3)

5.4 Give TWO safety rules for using electricity at home. (2)

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TOTAL: 75

END OF QUESTION PAPER